

Maxwell–Teleparallel Algebra & Chern–Simons Theory

Reference Notes

Lie Algebra Classification

Basic Definitions

Lower central series: $C^1\mathfrak{g} = \mathfrak{g}$, $C^{n+1}\mathfrak{g} = [\mathfrak{g}, C^n\mathfrak{g}]$; $\mathfrak{g}$ is *nilpotent* if $C^n\mathfrak{g} = 0$. **Derived series:** $D^{n+1}\mathfrak{g} = [D^n\mathfrak{g}, D^n\mathfrak{g}]$; $\mathfrak{g}$ is *solvable* if $D^n\mathfrak{g} = 0$. Nilpotent $\Rightarrow$ solvable. Radical $\text{rad}(\mathfrak{g}) =$ largest solvable ideal; $\mathfrak{g}$ is *semisimple* iff $\text{rad}(\mathfrak{g}) = 0$. Killing form $K(X, Y) = \text{Tr}(\text{ad}_X \text{ad}_Y)$; semisimple $\Leftrightarrow K$ nondegenerate (Cartan).

Lem 1.1. $[\text{ad}_x, \text{ad}_y] = \text{ad}_{[x, y]}$ (*Jacobi identity*). The center $Z(\mathfrak{g})$ is an ideal.

Cartan decomposition: $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$ with $[H, X_\alpha] = \alpha(H)X_\alpha$.

Free Lie Algebras

$\mathfrak{f}(X) \subset T(V)$ is the Lie subalgebra generated by X in the tensor algebra, where $[a, b] = a \otimes b - b \otimes a$. Universal property: every map $X \rightarrow \mathfrak{g}$ extends uniquely to $\tilde{\phi} : \mathfrak{f}(X) \rightarrow \mathfrak{g}$. By PBW, $T(V) \cong \mathcal{U}(\mathfrak{f}(X))$. Graded dimension:

$$\dim \mathfrak{f}(X)_n = \frac{1}{n} \sum_{d|n} \mu(n/d) |X|^d \quad (\text{necklace polynomial}).$$

$\text{rad}(\mathfrak{f}(X)) = 0$ for $|X| \geq 2$; every Lie algebra is $\mathfrak{f}(X)/\mathfrak{t}$.

Non-Semisimple Algebras

Solvable. Lie's theorem: representations are simultaneously upper-triangularizable over $\overline{\mathbb{F}}$. Prototype: Borel subalgebra $\mathfrak{b} \subset \mathfrak{sl}(n)$.

Nilpotent. Engel's theorem: $\mathfrak{g}$ nilpotent $\Leftrightarrow \text{ad } x$ nilpotent $\forall x$. Heisenberg algebra $\mathfrak{h}_{2n+1}$: $[p_i, q_i] = z$, two-step nilpotent.

Nilradical. $\text{nil}(\mathfrak{g}) \subseteq \text{rad}(\mathfrak{g}) \subseteq \mathfrak{g}$; for any Levi factor $\mathfrak{s}$: $[\text{rad}(\mathfrak{g}), \mathfrak{s}] \subset \text{nil}(\mathfrak{g})$.

Levi Decomposition

Thm 1.2 (Levi 1905). $\mathfrak{g} = \mathfrak{s} \ltimes \text{rad}(\mathfrak{g})$ for a semisimple Levi factor $\mathfrak{s}$; unique up to conjugation by $e^{\text{ad } x}$, $x \in \text{nil}(\mathfrak{g})$ (Malcev 1942).

Sketch. Induct on $\dim \text{rad}(\mathfrak{g})$. For abelian radical the extension is classified by $H^2(\mathfrak{s}, \text{rad}) = 0$ (Whitehead), hence splits. General case: reduce modulo $[\text{rad}, \text{rad}]$ and lift.

Universal Enveloping Algebra

$\mathcal{U}(\mathfrak{g}) = T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] \rangle$; left adjoint to $\mathcal{U} \rightarrow \text{LieAlg}$.

Thm 1.3 (PBW). Ordered monomials in a basis of $\mathfrak{g}$ form a basis of $\mathcal{U}(\mathfrak{g})$. Corollaries: $\iota : \mathfrak{g} \hookrightarrow \mathcal{U}(\mathfrak{g})$ injective; $\text{gr } \mathcal{U}(\mathfrak{g}) \cong S(\mathfrak{g})$; $\mathcal{U}(\mathfrak{g})$ is a Noetherian domain.

Poincaré & Maxwell Algebras

Poincaré: $[J_{ab}, J_{cd}] = \eta_{bc}J_{ad} - \dots$; $[J_{ab}, P_c] = \eta_{bc}P_a - \eta_{ac}P_b$; $[P_a, P_b] = 0$. Structure: $ISO(d-1, 1) = SO(d-1, 1) \ltimes \mathbb{R}^d$. Gauge connection $A = \frac{1}{2}\omega^{ab}J_{ab} + e^aP_a$; curvature $F = \frac{1}{2}R^{ab}J_{ab} + T^aP_a$.

Maxwell extension: Adjoin $Z_{ab} = -Z_{ba}$ with

$$[P_a, P_b] = Z_{ab}, \quad [J_{ab}, Z_{cd}] = \eta_{bc}Z_{ad} - \dots, \quad [P_a, Z_{bc}] = [Z_{ab}, Z_{cd}] =$$

Physical origin: particles in constant EM background, $Z_{ab} \sim F_{ab}$. Extended connection $A = \frac{1}{2}\omega^{ab}J_{ab} + e^aP_a + \frac{1}{2}k^{ab}Z_{ab}$; Z -sector curvature $F_{(Z)}^{ab} = dk^{ab} + e^a \wedge e^b$. Obtained from $\mathfrak{sp}(4)$ by İnönü–Wigner contraction.